SIMATS ENGINEERING
ASSESSMENT TOOL II — DATA ANALYSIS PROJECT

Course: Data Handling and Visualization	Code: DSA0603
Duration: 180 min	Total Marks: 100
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COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

Code of Conduct:

I Surya JK / 192324285 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Understanding of Concepts 25%	Analytical Design 25%	Use of Tools 20%	Analysis of Results 20%	Documentation 10%
Q1					
Q2					
Q3					
Q4					
Q5					
Total					

DSA0603 — DATA HANDLING AND VISUALIZATION

CO3_AT2 — Assessment Tool: DATA ANALYSIS PROJECT (with Model Answers)

Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by-Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

General Instruction: For each project task, use an appropriate visualization in R, justify the design choice, interpret the result in context, and identify potential visualization bias. R/ggplot2 (or an equivalent tool such as treemapify, ggalluvial, or networkD3) may be used. Submit R code, output plots, and a short written interpretation for each question.

Question 1 — Comparing Distributions with ECDF and Q-Q Plots

Scenario: Delivery time (hours) for two courier partners over 20 shipments each. Operations wants to know whether one partner is consistently faster and whether either partner's delivery times follow a normal pattern that would justify simple mean-based SLAs.

Partner	Delivery times (hours), n = 20
Partner A	18,19,20,20,21,21,22,22,23,23,24,24,25,26,27,28,30,33,38,46
Partner B	20,21,22,22,23,23,23,24,24,24,24,25,25,25,26,26,27,27,28,29

Answer

Summary statistics: Partner A — mean = 25.5 h, median = 23.5 h, sample SD ≈ 6.89 h, IQR = 6.5 (Q1=21, Q3=27.5). Partner B — mean = 24.4 h, median = 24 h, sample SD ≈ 2.33 h, IQR = 3 (Q1=23, Q3=26).

1. Overlaid ECDF plots

Plotting F_A(x) and F_B(x) on the same axes (x = delivery hours, y = cumulative proportion) shows Partner B's curve rising quickly and steeply to 1.0 by ~29 hours, while Partner A's curve rises more gradually and stretches out to 46 hours before reaching 1.0 — visually, Partner B's ECDF is compact and steep; Partner A's is elongated with a long, flat, low-slope tail on the right.

2. Normal Q-Q plots (each partner separately)

Partner B's Q-Q plot points fall close to the reference line across nearly the whole range — consistent with an approximately normal/symmetric distribution. Partner A's Q-Q plot follows the line reasonably well for the lower ~80% of points but curves sharply upward at the top end (30, 33, 38, 46 h), the classic signature of a right-skewed distribution with a heavy tail.

3. Point of maximum ECDF divergence and operational interpretation

Computing F_A(x) − F_B(x) at each threshold, the gap reaches its largest magnitude (≈0.20, i.e. a 20-percentage-point gap) in two places: around 21–22 hours, where F_A briefly leads F_B (Partner A completes a slightly larger share of its "fast" deliveries early), and again around 28–29 hours, where F_B leads F_A (Partner B has already delivered 100% of its shipments by 29 h, while Partner A has completed only about 80%).

Operationally, the second divergence (~29 h) is the more important one: it shows that although Partner A can be competitive or even faster on its best shipments, it carries a real risk of long-tail delays (30–46 h) that Partner B simply does not have — by the time Partner B has closed out every shipment, one in five of Partner A's shipments is still in transit.

4. Normality assessment and evidence of skew/outliers

Partner B: mean (24.4) ≈ median (24), Q-Q points hug the reference line, and by the 1.5×IQR rule (fences 18.5–30.5) every observation (20–29) falls inside the fences — the data reasonably supports a normal-distribution assumption, with no evidence of outliers.

Partner A: mean (25.5) noticeably exceeds median (23.5), the Q-Q plot's upper tail bows away from the line, and by the 1.5×IQR rule (fences 11.25–37.25) the values 38 and 46 exceed the upper fence and are statistical outliers, with 33 borderline. Partner A's distribution is right-skewed and does not support a normal assumption.

5. SLA recommendation

Empirical percentile-based SLA limits (e.g., the 90th or 95th percentile of actual delivery times) are recommended, at least for Partner A, since its skewed, outlier-containing distribution would make a mean ± SD limit both too loose in the middle of the distribution and misleading at the tail (mean + 2SD ≈ 39.3 h would still miss the 46-hour shipment while over-stating typical risk for the bulk of on-time deliveries). Partner B's near-normal, low-variance data would tolerate a mean ± SD limit reasonably well, but for a single, fair, comparable SLA policy across both partners, percentile-based limits are the safer and more defensible choice overall.

Question 2 — Distribution Shape Across Many Groups using Bean Plots

Scenario: Patient waiting time (minutes) in five outpatient departments over a week. Administration wants to identify which departments have consistent waiting times and which have unpredictable, multi-modal patterns before reallocating staff.

Department	Waiting times (minutes)
Cardiology	10,12,13,14,15,15,16,17,18,20
Orthopedics	8,9,25,26,27,9,10,28,29,8
Pediatrics	5,6,6,7,7,8,8,9,9,10
Dermatology	15,16,17,18,19,20,21,22,23,24
General Medicine	10,40,12,38,11,41,13,39,12,40

Answer

Descriptive statistics (n=10 each):

Department	Mean	Median	Range	Shape
Cardiology	15.0	15.0	10 (10–20)	Unimodal, roughly even spread
Orthopedics	17.9	17.5	21 (8–29)	Clearly bimodal: cluster ~8–10 and ~25–29
Pediatrics	7.5	7.5	5 (5–10)	Unimodal, very tight, low variability
Dermatology	19.5	19.5	9 (15–24)	Unimodal, evenly spread
General Medicine	25.6	25.5	31 (10–41)	Strongly bimodal: cluster ~10–13 and ~38–41

1. Bean-plot comparison

Placing all five departments along a shared vertical (minutes) axis, each with its mean line, jittered raw points, and a mirrored kernel-density curve, immediately distinguishes single-peak, symmetric-looking densities (Cardiology, Pediatrics, Dermatology) from densities with two distinct humps separated by a gap with almost no data (Orthopedics, General Medicine).

2. Departments with bimodal/multi-modal patterns

Orthopedics (values cluster tightly around 8–10 minutes and separately around 25–29 minutes, with nothing in between) and General Medicine (an even more extreme split: 10–13 minutes vs 38–41 minutes) both show clear bimodality. This pattern typically suggests two distinct patient/process sub-populations are being lumped together in the same average — e.g., quick, simple check-ins versus complex cases needing specialist review, walk-ins versus scheduled appointments, or a specific time-of-day/day-of-week bottleneck that only affects some patients.

3. Bean plot vs standard boxplot

A standard boxplot of Orthopedics or General Medicine would show a wide box/whisker span and a median line somewhere in the middle (e.g., ~17.5 or ~25.5), which looks like a single, moderately variable department. It conceals the fact that essentially no real patient actually waits close to that "typical" median value — every patient experiences either a short wait or a long wait, never a medium one. Only the bean plot's density curve reveals this two-cluster (bimodal) reality, which is critical information the boxplot's five-number summary hides.

4. Ranking by variability and by median

By variability (range): General Medicine (31) > Orthopedics (21) > Cardiology (10) > Dermatology (9) > Pediatrics (5).

By median waiting time: General Medicine (25.5) > Dermatology (19.5) > Orthopedics (17.5) > Cardiology (15.0) > Pediatrics (7.5).

The rankings differ because median (a measure of central tendency) and range (a measure of spread) capture different properties. Dermatology, for instance, ranks 2nd by median but only 4th by variability — it has a moderately high but consistent wait. Orthopedics, by contrast, ranks only 3rd by median yet 2nd by variability, because its bimodal split pulls the median toward the middle even though no patient actually experiences a "middle" wait — the median alone would understate how unpredictable Orthopedics really is.

5. Staffing recommendation

For Orthopedics and General Medicine, investigate and separate the two underlying patient streams (e.g., fast-track/simple-case slots versus dedicated specialist-review slots or blocks), since a single averaged staffing model is poorly suited to a bimodal demand pattern. Pediatrics is already fast and consistent and needs no change; Cardiology and Dermatology have single, moderate, predictable waits and can be staffed on straightforward average-demand planning.

Question 3 — Choosing an Honest Proportion Chart

Scenario: 300 customers across three store formats surveyed on preferred payment method. Senior management wants a dashboard chart that allows accurate comparison of payment-method share across formats without visually exaggerating small gaps.

Store Format	Cash (%)	Card (%)	UPI/Wallet (%)
Hypermarket	18	34	48
Supermarket	22	30	48
Convenience	35	25	40

Answer

1. Chart construction

Side-by-side (grouped) bar chart: for each store format, three adjacent bars (Cash, Card, UPI/Wallet) share a common zero baseline on one y-axis.

100% stacked bar chart: one bar per format, fixed total height of 100%, divided into three stacked segments proportional to Cash/Card/UPI share.

2. Three separate pie charts

One pie per store format (Hypermarket, Supermarket, Convenience), each divided into three wedges sized by Cash/Card/UPI percentage, drawn as three independent circles.

3. Ease of judging which format has the highest UPI/Wallet share

The side-by-side bar chart makes this easiest: all three UPI bars share the same zero baseline, so a viewer can see almost instantly that Hypermarket and Supermarket are tied at 48% and both exceed Convenience's 40%. The 100% stacked bar chart is harder, since the UPI segment sits at a different starting height in each bar (stacked on top of that format's Cash+Card total) and has no common baseline, making the small Hypermarket/Supermarket-vs-Convenience gap harder to read precisely. The three separate pie charts are hardest of all: the viewer must compare wedge angles across three different circles with no shared reference, and the 8-percentage-point gap (48% vs 40%) is easy to miss entirely.

4. Arc-length vs length perception

Per the Cleveland–McGill perceptual accuracy ranking, judging position/length along a common aligned scale (as in a bar chart) is one of the most accurate visual-comparison tasks humans can perform, while judging angle or arc length (as in a pie chart) is considerably less accurate — and accuracy drops further when the wedges being compared sit in separate charts rather than the same one. This is why the 8-point UPI gap is unmistakable as a bar-height difference but easy to overlook as an arc-length difference between three distinct pies.

5. A misleading design choice for Convenience's cash usage

Truncating the y-axis of a bar chart so it starts at, say, 15% instead of 0% would visually stretch the gap between Convenience's 35% cash share and the other formats' 18–22%, making Convenience look dramatically more cash-reliant than the real 13–17 percentage-point difference warrants. (Equally misleading alternatives: rendering the Convenience pie in 3D/tilted perspective, which enlarges front-facing wedges like Cash, or deliberately ordering/coloring the Convenience chart to draw the eye to the cash wedge while muting the other formats.) Any of these choices distorts the true magnitude of the difference and can lead management to over-react to a store format's cash dependency.

Question 4 — Nested Proportions in a Household Budget

Scenario: Average monthly household budget (₹60,000) aggregated by spending category and subcategory. The product team wants a dashboard visual showing both overall category share and subcategory composition within each category.

Category	Subcategory	Amount (₹)
Housing	Rent/EMI	18000
Housing	Utilities	4000
Housing	Maintenance	2000
Food	Groceries	9000
Food	Dining Out	3000
Transport	Fuel	3500
Transport	Public Transit	1500
Savings	Investments	10000
Savings	Emergency Fund	4000
Discretionary	Entertainment	3000
Discretionary	Shopping	2000

Answer

Category totals: Housing = 18000+4000+2000 = ₹24,000; Food = 9000+3000 = ₹12,000; Transport = 3500+1500 = ₹5,000; Savings = 10000+4000 = ₹14,000; Discretionary = 3000+2000 = ₹5,000. Sum = ₹60,000, matching the stated total household budget.

1–2. Treemap and sunburst construction

Treemap: five top-level rectangles (Housing, Food, Transport, Savings, Discretionary), area-proportional to category totals, each subdivided into nested rectangles for its subcategories (e.g., Housing splits into Rent/EMI, Utilities, Maintenance rectangles).

Sunburst: an inner ring of five arcs (one per category, arc length proportional to category total) and an outer ring of arcs for each subcategory, each positioned within its parent category's angular span, arc length proportional to subcategory amount.

3. Housing's share and its largest subcategory's share

Housing = ₹24,000 ÷ ₹60,000 = 40% of the total budget.

Housing's largest subcategory is Rent/EMI at ₹18,000, which is 18,000 ÷ 60,000 = 30% of the total budget (equivalently, 75% of the Housing category's own allocation).

4. Comparing two subcategories from different categories

The treemap makes this easier: all rectangles sit on the same flat plane, so two subcategories from different categories (e.g., Rent/EMI in Housing vs. Investments in Savings) can be compared directly by eye as two areas on the same 2-D surface, without needing to account for which ring or angular sector they belong to. In the sunburst, the same two subcategories sit in different angular positions (and possibly perceived at different effective radii within the outer ring), so the viewer must mentally normalize for position and curvature before comparing their arc lengths — a noticeably harder task.

5. A scenario where treemap area-encoding could mislead

Transport and Discretionary are exactly equal in the data (₹5,000 each, both 8.3% of the budget), but a squarified treemap layout algorithm can render equal-area rectangles with very different aspect ratios — e.g., Transport as a long thin strip along one edge and Discretionary as a more compact, near-square block. Even though both occupy identical area, the elongated shape is often perceived as smaller or less important than the compact shape, causing a viewer to wrongly conclude that Transport and Discretionary spending are unequal when they are, in fact, identical.

Question 5 — Flow Analysis using Sankey Diagram and Parallel Sets

Scenario: 250 students tracked from enrolled course, through device used to study, to final completion status. The academic team wants to understand dominant enrollment-to-outcome pathways and spot problem areas.

Course	Device	Completion Status	Students
Data Science	Mobile	Completed	20
Data Science	Mobile	Dropped	25
Data Science	Laptop	Completed	45
Data Science	Laptop	Dropped	10
Web Development	Mobile	Completed	15
Web Development	Mobile	Dropped	20
Web Development	Laptop	Completed	35
Web Development	Laptop	Dropped	10
Digital Marketing	Mobile	Completed	10
Digital Marketing	Mobile	Dropped	15
Digital Marketing	Laptop	Completed	20
Digital Marketing	Laptop	Dropped	25

Answer

1–2. Sankey diagram and parallel sets

Sankey diagram: three node columns — Course (Data Science, Web Development, Digital Marketing) → Device (Mobile, Laptop) → Completion Status (Completed, Dropped) — linked by curved paths whose width is proportional to student count (total flow = 250, confirmed by summing the table).

Parallel sets / alluvial plot: three parallel categorical axes (Course, Device, Completion Status) with ribbons connecting category values across axes, ribbon thickness proportional to student count, letting the same pathways be read as stacked segments at each axis.

3. Three largest flows and their percentages (of 250 total)

Rank	Flow	Students	% of total
1	Data Science → Laptop → Completed	45	18.0%
2	Web Development → Laptop → Completed	35	14.0%
3 (tie)	Data Science → Mobile → Dropped	25	10.0%
3 (tie)	Digital Marketing → Laptop → Dropped	25	10.0%

4. Weakest completion outcome and where drop-off is most visible

Completion rate by course: Data Science = (20+45)/100 = 65%. Web Development = (15+35)/80 = 62.5%. Digital Marketing = (10+20)/70 ≈ 42.9%.

Digital Marketing has by far the weakest completion outcome — it is the only course where dropped students (40) outnumber completed students (30). The drop-off is most visible at the Device → Completion Status stage: unlike the other two courses, where Laptop users complete far more than they drop (Data Science 45 vs 10; Web Development 35 vs 10), Digital Marketing's Laptop users still drop more than they complete (20 completed vs 25 dropped), and its Mobile users show the same pattern (10 completed vs 15 dropped). This indicates the problem is tied to the course content/engagement itself rather than to device access, since even the normally stronger "Laptop" pathway fails to convert for Digital Marketing.

5. How inconsistent ordering or unscaled link widths could mislead

If courses were ordered alphabetically rather than by enrollment or performance, Digital Marketing's weak outcome would not stand out visually and could be overlooked among the three. If link widths were not drawn strictly proportional to student counts (e.g., a minimum visible width enforced for small flows), a 10-student flow such as Data Science → Laptop → Dropped could appear almost as wide as the 45-student Data Science → Laptop → Completed flow, making Data Science look far less successful than it actually is and obscuring the real comparison — that Digital Marketing, not Data Science, is the course that most needs intervention.